Amrita School of Computing

23CSE311 Software Engineering

Lab Worksheet 2:

Software Requirements Specification

Date:

CO2	Apply requirement engineering principles to analyze, model, and validate requirements for effective software solutions.

Objective:

To clearly document all functional and non-functional requirements of the system, ensuring a shared understanding among stakeholders and serving as a reference for design, development, testing, and maintenance.

Team Information:

Team Member Name	Student ID	Role in Project
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Project Title: Real-Time campus companion

1.Introduction:

This project introduces a Real-Time campus companion designed to streamline communication, classroom allocation, and event updates within a college campus. The purpose of this application is to address common challenges faced by students, teachers, and campus staff, such as confusion over classroom availability, last-minute room changes, locating peers or faculty, and staying updated on college events.

The system allows students, teachers, class representatives (CRs), and even support staff to log in using their official college credentials. Teachers, CRs, and

campus workers can update classroom occupancy in real time, ensuring that students always know where their next class is being held. Notifications guide students whenever their classroom changes, while navigation support helps them reach new locations using campus landmarks.

In addition, the app provides features such as event listings with participation links, the ability to locate other sections' classrooms, and access to teacher availability and cabin details. By integrating these features into a single platform, the application serves as a personal campus assistant for every member of the college community.

The intended audience for this application includes:

- **Students: To track classes, events, and peers.**
- **Teachers & CRs: To update classroom status and share availability.**
- **Support Staff (Ayyammas/Workers): To help manage classroom status updates.**

Overall, the project aims to create a centralized, real-time, and user-friendly platform that improves campus coordination and reduces everyday confusion

1.1 Purpose

The purpose of the Real-Time campus campanion is to create a centralized digital platform that simplifies campus life by providing real-time updates, efficient classroom management, and seamless communication between students, faculty, class representatives, and support staff.

The app aims to:

- **Reduce confusion about classroom locations and availability.**
- **Ensure effective utilization of campus resources.**
- **Provide timely notifications to students about schedule or room changes.**
- **Enable easy access to college events, teacher availability, and peer classroom details.**
- **Act as a personal campus assistant to enhance the overall student and staff experience.**

Example Functionalities

1. **User Authentication**
    - **Students, teachers, CRs, and support staff log in using their official college email IDs.**
2. **Classroom Availability Management**

    - **Teachers, CRs, and support staff can mark classrooms as *occupied* or *free*.**
    - **Students can instantly check which rooms are free to avoid clashes.**
3. **Smart Notifications**
    - **Students receive notifications when their class is moved to a different room.**
    - **No alerts are sent if the class remains in the same room (to avoid spam).**
4. **Campus Navigation**
    - **Directions to classrooms or seminar halls using nearby landmarks.**
5. **Event & Activity Hub**
    - **List of upcoming events, fests, and seminars with participation/registration links.**
6. **Cross-Section Class Lookup**
    - **Students can find the real-time classroom of other sections by knowing their branch and section.**
7. **Teacher Availability & Cabins**
    - **Faculty cabin locations and availability for viva, mentoring, or consultations.**
8. **Time-Stamped Updates**
    - **Every classroom update shows the most recent edit time, ensuring accuracy.**

1.2 Document Conventions

This section defines the symbols, abbreviations, and notations used throughout the document to ensure clarity and consistency.

- **UI** – User Interface
- **UX** – User Experience
- **OTP** – One-Time Password
- **API** – Application Programming Interface
- **DB** – Database
- **SQL** – Structured Query Language
- **GPS** – Global Positioning System
- **CR** – Class Representative
- **ID** – Identification

- **App** – Application
- **Noti.** – Notification
- **Auth** – Authentication

1.3 Intended Audience and Reading Suggestions

This document is intended for all stakeholders involved in the development, testing, deployment, and usage of the College Student Management Application. Each group can focus on the sections most relevant to their role:

- **Developers** – Use this document to implement application features, database integration, and ensure the system architecture aligns with requirements.
- **QA Testers** – Validate that the application functions correctly, meets performance benchmarks, and ensures security across all modules.
- **Business Analysts / Project Leads** – Ensure that the application aligns with institutional needs, solves real campus challenges, and delivers value to all user groups.
- **Students** – Understand how to use the app to track classes, receive notifications, and access events.
- **Teachers & Class Representatives (CRs)** – Learn how to manage classroom updates, share availability, and guide students effectively.
- **Support Staff (Ayyammas / Workers)** – Understand their role in updating classroom availability within the app.
- **Administrators** – Oversee overall system usage, user management, and ensure smooth adoption of the app across the campus.

Reading Suggestions:

- Technical readers (Developers, Testers) should focus on the **System Design, Functional Requirements, and Use Case Diagrams**.
- Non-technical readers (Students, Teachers, Staff) should focus on the **User Guide and Functionalities section** to understand usage.
- Analysts and Administrators should review the **Objectives, System Features, and Limitations** for strategic understanding.

1.4 Product Scope

The **Real-Time campus companion** is a high-level digital solution designed to improve coordination and communication within a college campus. The application provides a centralized platform where students, teachers, class representatives, and support staff can efficiently manage classroom availability, track events, and stay updated on schedules in real time.

Key Capabilities:

- **Classroom Management:** Teachers, CRs, and support staff can update whether classrooms are free or occupied, with time-stamped updates for accuracy.
- **Real-Time Notifications:** Students receive alerts when their classroom changes, ensuring they reach the correct location without confusion.

- **Navigation Support:** Built-in campus navigation using nearby landmarks helps users easily find classrooms, seminar halls, and teacher cabins.
- **Event Management:** Displays upcoming college events, seminars, and fests with direct participation/registration links.
- **Cross-Section Classroom Lookup:** Enables students to locate classrooms of peers in other branches and sections.
- **Teacher Availability:** Provides information about faculty availability and their cabin locations for viva, project discussions, or mentoring.
- **User-Friendly Access:** Login via official college email ensures secure access for students, faculty, CRs, and support staff.

1.5 References

The following tools, resources, and standards were used during the preparation of this document and the planning of the College Student Management Application:

- **Google Docs** – For collaborative document drafting and editing.
- **Notion** – For structured project planning, requirement tracking, and note organization.
- **Microsoft OneNote** – For quick brainstorming, capturing ideas, and managing meeting notes.
- **IEEE SRS Standards (IEEE 830-1998)** – For guidelines in structuring the Software Requirements Specification document.
- **Agile Methodology Documentation** – For aligning development with iterative and incremental practices.
- **UML (Unified Modeling Language)** – For preparing system diagrams and visual representations of workflows.
- **Database Design References (SQL Documentation & Tutorials)** – For schema design and data management.
- **Android Developer Documentation / Flutter Docs** – For mobile app design and implementation guidelines.
- **API Documentation Standards (Swagger / Postman)** – For designing and testing APIs used in the application.

2. Overall Description

This section provides a general overview of the software's functionality.

2.1 Product Perspective

The **Real-Time campus campanion** is envisioned as a **multi-user campus management platform** that integrates students, teachers, class representatives (CRs), and support staff into a single digital ecosystem. Unlike traditional static timetables or

notice boards, this system provides **real-time updates** on classroom occupancy, schedules, and events.

The application does not replace existing institutional systems but rather **enhances them** by leveraging:

- **Official College Email Authentication** – ensures secure access for only authorized users.
- **Timetable Data Integration** – imports and aligns classroom allocations with real schedules.
- **GPS & Campus Landmarks** – assists students in navigating classrooms and event halls within the campus.
- **Push Notifications** – keeps students instantly informed about classroom changes or upcoming events.

The app fits into the existing college environment as a **centralized assistant**, bridging gaps between students, faculty, CRs, and support staff to improve resource utilization and reduce confusion across the campus.

2.2 Product Features

The core functionalities of the Real-Time campus campanion include:

- **User Registration & Login** – Secure login via official college email IDs to ensure only authenticated users can access the app.
- **Classroom Availability Management** – Teachers, Class Representatives (CRs), and support staff can update whether a classroom is free or occupied, along with a time-stamp.
- **Real-Time Notifications** – Students receive instant alerts if their class is relocated to another room, preventing confusion.
- **Classroom Navigation** – Students can access directions to classrooms and event halls using GPS and campus landmarks.
- **Event Updates & Participation** – Displays details of upcoming college events, seminars, and fests with direct registration or participation links.
- **Cross-Section Classroom Lookup** – Enables students to locate the real-time classroom of peers in other branches or sections by simply entering branch and section details.
- **Teacher Availability & Cabin Details** – Shows faculty availability status and cabin locations for viva, mentoring, or consultations.
- **Timetable Integration** – Imports official timetables to automatically align classroom schedules and reduce errors.
- **Admin Dashboard** – Provides administrators with control over user management, system monitoring, and overall data insights

2.3 User Classes and Characteristics

The Real-Time campus campanion supports multiple user groups, each with specific roles and responsibilities:

- **Students**
  - Primary users of the application.
  - View classroom availability and receive real-time notifications about room changes.
  - Navigate to classrooms, seminar halls, and events using campus landmarks.
  - Access event details and register for participation.
  - Look up classrooms of other sections (using branch & section info).
- **Teachers**
  - Update classroom availability and mark rooms as occupied/free.
  - Share their availability for viva, mentoring, or extra sessions.
  - Provide information about their cabin location.
- **Class Representatives (CRs)**
  - Responsible for frequently updating classroom status on behalf of their section.
  - Ensure accuracy of timetable alignment with real-time usage.
  - Assist teachers in communicating schedule or room changes.
- **Support Staff (Ayyammas / Workers)**
  - Update classroom availability status when rooms are cleaned, closed, or unused.
  - Provide backend support for keeping real-time room information accurate.
- **Administrators**
  - Manage user access and permissions across different roles.
  - Oversee system usage, event updates, and data accuracy.
  - Monitor performance and ensure smooth functioning of the application.

2.4 Operating Environment

The Real-Time campus campanion will run on commonly used mobile and web platforms, ensuring accessibility for all stakeholders. The system requirements are as follows:

- **Mobile Application:**

    - Android 9.0 (Pie) or higher
    - iOS 13.0 or higher
- **Web Application (Admin Dashboard):**
    - Supported Browsers: Google Chrome, Mozilla Firefox, Safari, Microsoft Edge
    - Minimum Resolution: 1280x720
- **Development Tools:**
    - **IDE/Editor:** Visual Studio Code / Android Studio
    - **Version Control:** Git & GitHub
    - **API Testing:** Postman
    - **Design & Documentation:** Figma, Notion, Google Docs
- **Backend & Database:**
    - **Database:** PostgreSQL / Firebase (for real-time updates & authentication)
    - **Server-Side Framework:** Node.js + Express.js (or Firebase Functions)
    - **Authentication:** Google OAuth (via college email IDs)
- **Cloud Hosting & Deployment:**
    - AWS / Google Cloud / Microsoft Azure (for backend & database hosting)
    - Firebase Cloud Messaging (for real-time notifications)
- **Navigation Support:**

2.5 Design and Implementation Constraints

The Real-Time campus campanion Application must adhere to the following design and implementation constraints to ensure reliability, scalability, and security:

- **User Load Handling**
    - The system should support at least **5,000 concurrent users** during peak usage (e.g., exam preparation days).
- **Data Security**
    - All uploaded files (e.g., notes, PPTs, question papers) must be **encrypted using AES-256** before storage.
    - User authentication and authorization must be implemented using **JWT tokens** for secure access.
- **Cross-Platform Compatibility**
    - The platform must run smoothly on modern browsers (**Chrome, Firefox, Edge, Safari**) and be fully mobile-friendly.
    - The mobile app should function efficiently on **Android 9+** and **iOS 13+** devices.

- **File Management**
  - File uploads should not exceed **50 MB per file**, considering bandwidth and storage limitations.
- **Cloud Deployment**
  - The system should be deployable on **budget-friendly cloud hosting providers** (AWS, Google Cloud, Firebase, or Azure) with **auto-scaling options**.
- **Access Control**
  - Branch-specific access must be enforced using **student roll numbers**, preventing unauthorized cross-branch/class data access.
- **Performance**
  - The **average response time** for accessing classrooms, files, or updating the tracker should not exceed **2 seconds under normal load**.
  - APIs must follow **RESTful standards** and be lightweight, optimized for **low-bandwidth mobile connections**.
- **Development Standards**
  - The system must be developed following the **Agile methodology** with iterative sprints.
  - **Version control** should be maintained using **Git + GitHub** for collaboration and tracking changes.
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2.6 Assumptions and Dependencies

Users

- Students and teachers will have a **stable internet connection** to upload, download, and access materials.
- Students will log in using **valid roll numbers and official college email IDs** to ensure branch-specific access.
- Students are motivated to use the **progress tracker and leaderboard features** actively to encourage participation and collaboration.

System

- The **server clock** is accurate to support countdown timers, auto-release of study materials, and scheduling features.
- The chosen database (**PostgreSQL or Firebase**) will remain **stable and reliable** for data storage and retrieval.
- The platform will face **moderate usage** during normal weeks, with **high load surges only during exam seasons**.

Dependencies

- Dependence on **faculty** to upload official materials (PPTs, instructions, question banks) in a timely manner.
- Dependence on **students** to contribute notes and past question papers to keep the leaderboard and knowledge-sharing features engaging.
- Dependence on **third-party services** such as:
  - **Firebase** – for storage, authentication, and notifications.
  - **GitHub** – for version control and collaborative development.
  - **Node.js libraries** – for backend processing and API handling.

3. Specific Requirements

3.1 Functional Requirements

3.1 User Registration & Authentication

- Users must log in using their **official college email ID and roll number**.
- Authentication is handled via **Google OAuth** or college-provided credentials.
- Only authorized users (students, teachers, CRs, staff, and admins) can access the app.
- Password reset and account recovery must be supported.

3.2 Classroom Availability Management

- Teachers, CRs, and support staff can **update classroom status** (Free/Occupied).
- Every update must include a **time-stamp** for accuracy.
- Students can **search for available classrooms** in real-time.
- Admin can monitor changes and prevent misuse (false updates).

3.3 Real-Time Notifications

- Students receive **push notifications** when their class is relocated.
- Notifications should only trigger if the classroom changes, avoiding spam.
- Admin can send **broadcast notifications** for important announcements.

3.4 Classroom Navigation

- The system provides **directions to classrooms** and event halls using **GPS + campus landmarks**.
- Users can view a **map of the campus** with key locations marked.
- Navigation should work even with low-bandwidth connections.

3.5 Event Management & Participation

- Displays **upcoming events, seminars, and fests** in a calendar view.
- Students can **register for events** using in-app links.
- Event organizers (teachers/admins) can update schedules and details.

3.6 Cross-Section Classroom Lookup

- Students can enter a **branch + section** to find the real-time classroom of peers.
- Lookup should only be valid during active class hours.
- Prevents misuse by restricting unnecessary access to other sections.

3.7 Teacher Availability & Cabin Details

- Teachers can set their **availability status** (Available/Busy).
- The system displays **faculty cabin numbers** for easy navigation.

- Students can request **appointments or viva slots** via the app.

3.8 Timetable Integration

- Imports the official **college timetable** for all sections.
- Automatically aligns classroom schedules with real-time usage.
- Allows manual corrections by CRs/teachers if sudden changes occur.

3.9 Admin Features

- Admin manages **user roles, access rights, and permissions**.
- Admin can monitor system performance and generate **usage reports**.
- Admin ensures compliance with **data security and access policies**.

3.2 External Interface Requirements

User Interfaces

- Mobile app (Android 9+ / iOS 13+) with **clean and intuitive design**, supporting both **dark and light modes**.
- Web app (Chrome, Firefox, Safari, Edge) for teachers and admin staff with responsive UI.
- Accessible **dashboard for admins** with real-time data visualization.

Hardware Interfaces

- GPS-enabled navigation support for **campus directions**.
- Mobile device camera integration for **QR code scanning** (for event check-ins or classroom verification).
- Devices must have a stable internet connection (Wi-Fi or 4G/5G).

Software Interfaces

- **Google Maps API** (for classroom/event hall navigation).
- **Firebase / PostgreSQL** (for authentication, file storage, and real-time database).
- **GitHub / GitLab** (for version control).
- **Payment Gateway (optional)** if event fees or registrations are needed.

Communication Interfaces

- **Push notifications** for classroom changes, events, and reminders.
- **Email alerts** for official announcements.
- **In-app chat/announcement board** for teacher-student communication.

3.3 Performance Requirements

- The app must handle **up to 5,000 concurrent student logins** during peak hours (exam schedules or major events).
- Classroom availability updates must reflect in **real-time (<2 seconds delay)** across all users.
- Push notifications should be delivered within **3 seconds** of the event trigger.
- The navigation system should update GPS location with a delay of no more than **5 seconds**.
- The platform must support **storage of at least 50,000 classroom/event records** and **100,000 student activity logs**.
- Average **API response time** should not exceed **2 seconds** under normal load.
- The system should remain stable with a **99.5% uptime** guarantee.

3.3 Performance Requirements

 The system must support **up to 5,000 concurrent users** during peak usage (exam schedules, seminar events).

 Classroom availability updates should reflect across the platform in **less than 2 seconds**.

 Event reminders and push notifications must be delivered within **3 seconds** of being triggered.

 The app should maintain **real-time responsiveness** with API response times under **2 seconds** in normal load conditions.

 GPS-based **classroom or seminar hall navigation** should update location data within **5 seconds**.

 The system must be scalable to handle:

- **50,000+ classroom/event records**
- **100,000+ student activity logs**
- **Daily uploads of at least 10,000 files** (notes, question papers, PPTs).

 The platform should ensure **99.5% uptime** to guarantee availability during academic peak times.

 The mobile and web applications must load the **home dashboard within 3 seconds** on a stable internet connection.

3.4 Security Requirements

 User Authentication & Authorization

- All users must log in with valid credentials (college email/roll number).
- Authentication will be implemented using **JWT tokens** with session expiration.

- Role-based access control (RBAC) must be enforced (Students, Faculty, Admin, Delivery Partners).

 Data Protection

- All sensitive data (notes, PPTs, question papers) must be encrypted using **AES-256** before storage.
- Passwords must be hashed using **bcrypt or PBKDF2** before saving in the database.
- User personal information (phone numbers, email addresses) must be protected and never exposed to unauthorized parties.

 Secure Communication

- All communication between client and server must use **HTTPS with TLS 1.2+**.
- API endpoints must implement **rate limiting** and input validation to prevent brute-force attacks and injection vulnerabilities.

 File & Access Security

- Uploaded files must be scanned for viruses/malware before storage.
- File access must be **branch-specific**, ensuring students cannot view or download materials from other branches.
- File size is restricted to **50 MB per upload** to minimize risks and ensure bandwidth efficiency.

 Session & Data Privacy

- User sessions will auto-expire after a period of inactivity (e.g., 15 minutes).
- Location-based features (navigation within campus) will request explicit **user consent** before enabling GPS.
- Logs of login attempts and file access will be stored for **audit and security monitoring**.

 Compliance & Backup

- The system must comply with **data protection guidelines** (such as GDPR principles for academic data).
- Regular backups of data and logs must be taken and securely stored on cloud infrastructure.

3.5 Software Quality Attributes

 Availability

- The platform must maintain **99.9% uptime**, ensuring students and teachers can access materials without interruption, especially during peak exam seasons.

 Scalability

- The system should **scale horizontally** to handle at least **5,000+ concurrent users** during peak load.
- Auto-scaling on cloud infrastructure must be enabled to adapt to fluctuating traffic.

 Usability

- The **User Interface (UI)** should be intuitive, mobile-friendly, and require minimal clicks for key tasks (upload, download, progress tracking).
- Support for both **dark and light modes** should be available.

 Reliability

- Data (notes, question papers, trackers) must be stored securely with **regular backups** to prevent loss.
- The system must recover gracefully from server crashes within **5 minutes**.

 Performance

- File uploads/downloads should complete within **2–3 seconds** under normal network conditions.
- Progress tracker updates must reflect within **1 second** of user action.

 Security

- Must comply with best practices for encryption, authentication, and role-based access.
- Regular **penetration testing** should be conducted to identify vulnerabilities.

 Maintainability

- Codebase should follow **modular and clean coding standards** to allow easy updates.
- Documentation (API docs, architecture diagrams) must be updated regularly.

 Portability

- The system should run on **modern browsers (Chrome, Firefox, Edge, Safari)** and be deployable across **Android/iOS mobile apps** with minimal modifications.

 Accessibility

- The platform should adhere to **WCAG 2.1 guidelines**, ensuring accessibility for users with disabilities (e.g., screen readers, color contrast).

3.6 Other Non-Functional Requirements

 Legal Compliance

- The system must comply with **GDPR** for data protection and privacy.
- For payment processing, compliance with **PCI-DSS standards** is mandatory.

- User-generated content (notes, question papers) must adhere to **copyright and academic integrity guidelines**.

 Localization & Internationalization

- Support for **multiple languages** (English, Hindi, Telugu, etc.) to cater to diverse student populations.
- Currency localization for payments (e.g., INR, USD, EUR) in case of global usage.
- Date, time, and number formats should adjust based on regional preferences.

 Accessibility

- The platform should comply with **WCAG 2.1 accessibility standards**.
- Features like **screen reader support, high-contrast mode, and voice commands** must be provided for visually impaired users.
- Keyboard-only navigation should be fully supported.

 Interoperability

- The system must integrate seamlessly with **third-party services** such as Firebase (storage), GitHub (version control), Google Maps API (location services), and payment gateways.
- Standardized **RESTful APIs** should be provided for future system expansions.

 Audit & Monitoring

- Maintain detailed **logs of user activities** (logins, uploads, downloads, updates) for security audits.
- Implement **real-time monitoring** and alert mechanisms for performance and security issues.

 Disaster Recovery & Backup

- Daily **automated backups** of all critical data must be maintained.
- A disaster recovery plan must ensure system restoration within **1 hour** in case of catastrophic failure.

 Ethical Usage

- The platform must prevent misuse, such as **plagiarism, sharing of irrelevant or harmful content**, and ensure data sharing remains within academic boundaries.

Instructor's Signature & Comments (For Lab Use):